neutral investors set λ to be small. Setting lambda to be zero is equivalent to our first criterion—minimize cost—since the risk term would be ignored.

This trading goal is also known as the standard "cost-risk" optimization or algorithmic optimization objective function. It is formulated as follows:

$$Min\ \mathrm{Cost}' + \lambda \cdot \mathrm{Risk}' \tag{8.21}$$

Example: An investor looking to minimize the combination of market impact cost (not including price appreciation) and timing risk for a specified risk aversion value λ will determine the optimal trading rate through minimizing the following equation:

$$Min\colon (b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I') + \lambda \cdot \left(\sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1}} \cdot P_0 \right) \tag{8.22}$$

Solving for the optimal trading rate we get:

$$\alpha^* = \left(\frac{b_1 \cdot I}{\lambda \cdot \sigma \cdot P_0 \cdot \sqrt{\frac{1}{3} \cdot \frac{1}{250} \cdot \frac{X}{ADV}}} \right)^{\frac{2}{3}} \tag{8.23}$$

Notice that this optimal solution is in terms of the investor's risk aversion parameter.

Figure 8.2d illustrates the BES for an investor with a risk aversion $\lambda = 1$. In this case the solution is at the point where the tangent to the ETF is -1. This is noted by strategy A4 on the ETF and has cost = \$0.28/share and timing risk = \$0.25. The trading rate that achieves this optimal strategy is 57% and corresponds to POV = 36%.

5. Price Improvement

The fifth criterion "Price Improvement" is used by investors wishing to maximize the probability that they will execute more favorably than a specified cost. Usually, this is the goal of participants seeking to maximize short-term returns or exploit a pricing discrepancy. Additionally, it is often the goal used by agency traders seeking to maximize the likelihood of outperforming a cost such as a principal bid, or the strategy utilized by a principal trading desk looking to minimize chances of gamblers ruin and maximize profiting opportunity. The proof of the price improvement strategy was derived by Roberto Malamut (see *Optimal Trading Strategies,* p. 225, and Kissell, Glantz, and Malamut (2004, p 45).